

Ladder Safety Assessment Using Head-mounted 360-degree Camera-based Posture Estimation Overlayed Real-time in Augmented Reality

Hari Iyer*
Arizona State University

Josiane Isingizwe†
Arizona State University

Ricardo Eiris‡
Arizona State University

Heejin Jeong§
Arizona State University

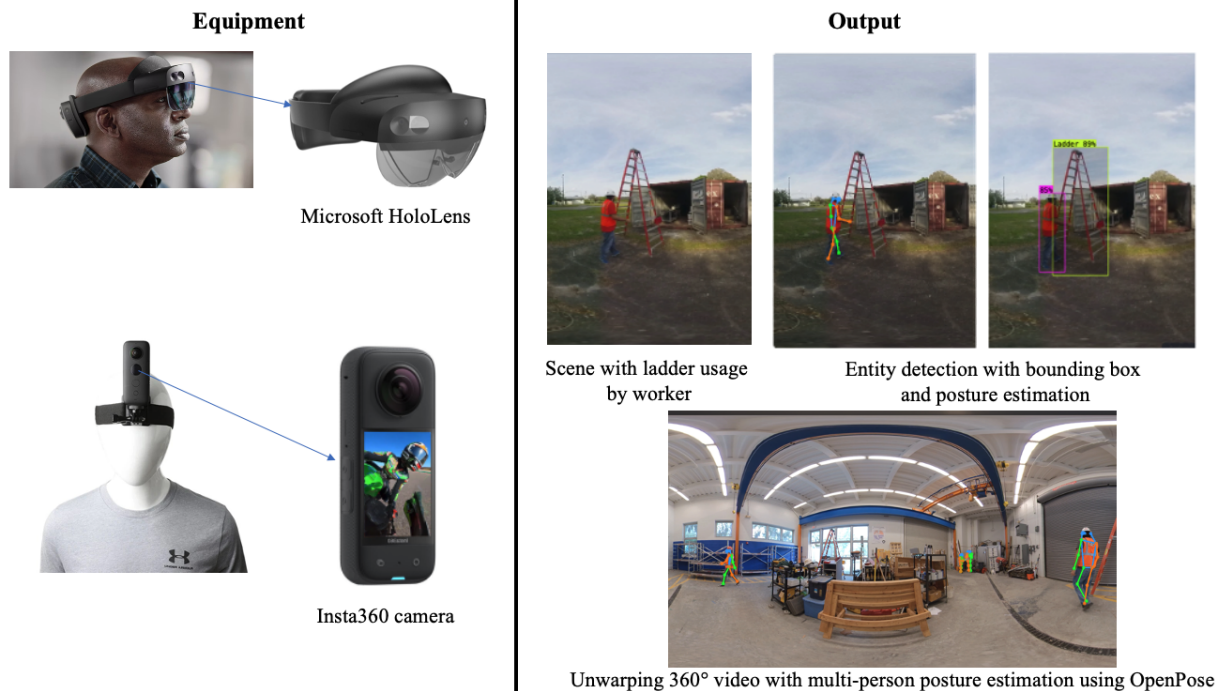


Figure 1: Proposed architecture for the implementation of the AR system and 360-degree camera for posture estimation.

ABSTRACT

Various industries involve overhead tasks using ladders, presenting potential risks of imbalance and falls, which are crucial factors impacting worker safety. This study proposes employing a 360-degree camera with augmented reality (AR) headset-driven imagery to conduct posture estimation for multiple on-site ladder workers. By utilizing posture estimation and identifying factors contributing to fall risks, the study aims to monitor worker movements to improve safety and task performance. The proposed system entails deploying 360-degree cameras mounted on an ergonomist's headgear, who monitors the workplace while wearing an AR headset. The head-mounted camera captures comprehensive footage, while the ergonomist views the estimated posture overlayed real-time in AR. Posture estimation models assess workers' joint movements, generating scores based on the Assessment of Repetitive Tasks (ART) metric. This system also assesses and generates feedback concerning workers' use of personal protective equipment. This proposed solution can potentially improve ladder safety and advance AR technology in occupational

*e-mail: hari.iyer@asu.edu

†e-mail: jisingiz@asu.edu

‡e-mail: reiris@asu.edu

§Corresponding author, e-mail: heejin.jeong@asu.edu

safety.

Index Terms: Assessment of repetitive tasks—Augmented reality (AR)—Occupation safety—Posture estimation

1 INTRODUCTION

The modern demands of industrial work rely on the safety and efficiency of tasks performed at elevated heights, especially those involving ladder use. Falls that occur due to slips and trips present serious risks to workers across various sectors, often influenced by multifaceted factors encompassing human behavior, environmental conditions, and task nature [13]. Ladder-related accidents continue to be a major concern, leading to severe injuries and productivity losses. Individual attributes, task specifics, and workplace conditions significantly influence an individual's ability to prevent falls and regain balance [15]. In 2020, there were more than 22,000 ladder-related injuries, and work domains focusing on installation, maintenance, and repair accounted for more than 5,000 ladder-related reported injuries [7]. Construction work is an example domain wherein ladder is used for reaching areas at a height. Chi et al. [9] investigated fatal falls in the construction industry, identifying contributing factors including characteristics of individual nature, site of fall, size of the company, and factors that led to the falls. Individual factors, like age, gender, experience, and adherence to personal protective equipment (PPE), significantly influenced fall outcomes. To address ladder safety concerns (e.g., falls, overloading, imbalance), the proposed study aims to utilize human-in-the-loop analysis, monitoring worker posture through quantifiable posture estimation

with 360-degree imagery integrated with Augmented Reality (AR). A head-mounted camera is used to capture the 360-degree imagery, as it has been widely used across applications [4, 10, 23]. Spherical imagery can be processed using 360-degree imagery [27], which facilitates the proposed experiment to obtain posture from all the ladder workers on the scene.

4 Stages of ART				
Stage A	A1: Arm Movement		A2: Repetition	
Frequency and Repetition				
Stage B	B: Force exerted with the hand			
Force				
Stage C	C1:	C2: Back	C3: Arm	C4:
Awkward Postures	Head/Neck Posture	Posture	Posture	Wrist/hand/ finger Posture
Stage D	D1:	D2:	D3:	
Additional Factors	Breaks	Work Pace	Duration	

Figure 2: ART's four stages (A, B, C, D) with their body parts and features of interest for awkward posture assessment.

The proposed study also involves analyzing muscle activation patterns using electromyography (EMG) sensors during ladder ascent and descent. This analysis identifies task-specific muscle groups prone to fatigue, assisting in ladder recalibration or repositioning. Milligan et al. [17] found that certain techniques of ladder climbing and corresponding experience alleviate physiological workload while also producing the desired outcome. Analyzing muscular demand involved in ladder operations personalize ladder settings, reducing accidents due to overexertion and enhancing work efficiency, thereby lowering the risk of falls. We aim to utilize the Assessment of Repetitive Tasks (ART) tool (see Figure 2) to assign scores to workers' posture during ladder ascent and descent tasks, which may involve repetitive or awkward upper limb motions. Probable risk of work-related musculoskeletal disorders (WMSDs) can change based on the demand, frequency, and time duration of such exposure during action performance, quantitative assessment enables ergonomists to select tasks with a elevated risk. This detection allows for immediate ergonomic intervention to prevent WMSDs [12]. Additionally, continuous real-time monitoring of heart rate and its variability via electrocardiogram (ECG) sensors during ladder-related tasks quantify cardiovascular stress imposed by ladder operations. The data, along with demographic information (e.g., age, gender) and experience levels, provides insights into the physiological stress associated with ladder workers. Integrating data sources, such as posture estimation, AR overlay, EMG, and ECG drives software-driven decision and feedback systems, enabling informed decisions about ladder settings to enhance fall safety.

This project has two primary objectives: (1) proposes an experimental and analysis framework for estimated worker posture and sensor-based data using machine learning techniques to assess fall risk during ladder use, and (2) outline testing procedures for PPE among ladder workers to improve Assessment of Repetitive Tasks (ART) scores. These objectives are built to examine the hypothesis: Head-mounted 360-degree camera-based posture estimation overlaid real-time in AR provides a more accurate and rapid assessment compared to the traditional method using a stationary 360-degree camera.

The proposed solution not only contributes to a safer working environment but also fosters a proactive safety culture by providing workers with tools for self-correction and adherence to safety guidelines during ladder-related tasks. The potential applications of this innovative approach extend to advancements in AR-enhanced occupational safety for other repetitive or high-risk work.

2 RELATED WORK

Ladder work-related falls are the second most frequent cause for deaths in the construction industry of US [14]. Preemptive measures

to detect fall risk can help mitigate the risk for such incidents. Barnett et al. [3] conducted a study to evaluate the safety of A-Frame ladders for certain overhead duct-based tasks. In the study, it was found that the forces linked to the installation of typical ductwork sometimes exceeded the lateral resistance of specialized A-Frame ladders. To detect these risks, a multi-dimensional approach involving objective sensor data and posture analysis can be used. The impact of ladder-related falls on the emergency department (ED) and recommendations for ladder safety were studied by [8]. It was concluded that ladder-related falls result in considerable workload to the ED. The study made recommendations for improving safety through interventions aimed at ladder users highest at risk of falls which are men, over 50 years old who performing domestic tasks. Safety interventions focused on potential task avoidance, training, use of safety equipment and suitable ladder setup. Out of these interventions, the current study focuses on utilisation of safety equipment in the form of PPE, and analyzes it with biomedical sensors. Panoramic AR has been used to develop a virtual environment for safety training at construction sites by [11]. They designed suitable augmentations in the interlinked places to show safety information regarding ladder fall hazards. In the currently proposed study, a 360-degree camera will be head-mounted on the ergonomist who will also wear an AR headset to overlay posture on the workers in the field of view. Previous studies [4, 10, 26] have used 360-degree imagery to analyze scenes for different applications including object detection and construction safety. There could be multiple workers at site that conduct ladder work, and the 360-degree imagery can help include all the workers for their posture analysis.

3 PROPOSED ARCHITECTURE

This study aims to use Insta360's 360-degree imagery mounted on a professional ergonomist's head, who wears the Microsoft HoloLens headset. Figure 1 shows the architectural pipeline. Once the 360-degree imagery is captured, posture estimation using OpenPose overlays data onto visible workers in AR through the HoloLens camera (Microsoft Corporation, www.microsoft.com/en-us/hololens), using Bluetooth or Wi-Fi-driven data transfer. Additionally, objects in the frame (e.g., ladder, PPE) are identified and isolated using a bounding box technique. This process assists the ergonomist in real-time assessment of ART scores for the workers during ladder ascent, descent, and movement tasks while gathering EMG and heart rate information.

The analysis would begin by unwarping and converting 360-degree footage into flattened frames that panoramically capture the entire field of view from the Insta360 camera (Arashi Vision Inc., www.insta360.com), encompassing all workers for analysis. Estimated ART scores, along with validation data from sensors capturing muscular and cardiac activity, is used to assess fall risk and identify awkward ergonomic posture in workers. This analysis aims to propose improvements for safer task execution.

4 PROPOSED EXPERIMENTAL SETUP AND ANALYSIS FRAMEWORK

We aim to recruit 24 ladder workers through flyers and social media. Participants will undergo screening based on specific criteria: age (at least 18 years old), physical capability to ascend and descend ladders, and no recent injuries within the past three years. During the experiment, participants will ascend and descend ladders while equipped with PPE including anti-slip gloves (e.g., Impacto: BG408, AV408) and anti-slip foot protection (e.g., Impacto SLKGRP). The tasks utilize OSHA- and ANSI-certified ladders, such as extension ladders and A-frame step ladders.

The proposed experiment employs a 2×2 design with four combinations of PPE wear: (1) gloves and foot protection worn, (2) gloves worn without foot protection, (3) no gloves worn but with foot protection, and (4) gloves and foot protection both not worn.



Figure 3: (a) Sample EMG (top) and IMU (bottom) data for flexor carpi ulnaris muscles for a participant's right arm performing a representative physical task (Delsys Inc.). (b) Heart rate data for the participant while performing the same task (Polar Electro).

For within-subject analysis, each participant will perform ladder ascend and descend tasks for all four conditions, while collecting all the four kinds of data (posture estimation, AR overlay, EMG, and ECG). The Trigno EMG system that measures maximum muscle contraction and activation patterns will be used, focusing on frequently used limb and trunk muscles, such as flexor digitorum profundus, flexor pollicis longus, and extensor digitorum communis (Delsys Inc., USA, www.delsys.com). Figure 3 shows sample output of the EMG and corresponding Inertial Measurement Unit (IMU) readings for flexor carpi ulnaris muscle for a representative physical task. IMU data provides acceleration and angular movement of the muscle that the sensor is mounted on. This data can be used for validation of the muscle activation recorded. Four EMG sensors are placed, one on each leg on the quadriceps vastus lateralis [21] and calves gastrocnemius [17]. For the arms and shoulders, two sensors with the channels will be employed to capture biceps and triceps on the biceps brachii muscle [2]. On the back, a sensor will be placed on each side of the flank on the latissimus dorsi muscle [22]. Two sensors will be placed on the forearms' flexor carpi ulnaris muscles [18]. Polar H10 Heart Rate sensors monitor heart rate and cardiovascular response (Polar Electro, Kempele, Finland, www.polar.com). The study will record default ladder settings, including inclination angle, ladder height, and rung adjustment, and assess deviations from

ideal positions based on sensor data. Analyses will be leveraged to establish relationships between muscle activation, ART scores from estimated posture, and cardiovascular responses. Additionally, sensor data will be used to compare ladder settings and suggest improvements, providing comprehensive insights into biomechanical and physiological aspects of ladder-related activities.

Analyzing ECG and EMG data will involve employing various statistical techniques. For ECG data, we aim to include heart rate variability [20, 24, 25] and power spectral analysis [16]. There was a survey conducted for heartbeat classification for detecting arrhythmia using ECG [1]. Similarly, we aim to apply statistical approaches for arrhythmia detection employing signal processing techniques like filtering and wavelet analysis. Machine learning techniques like expected projection of muscle activity will play a role in the classification of cardiac conditions based on ECG patterns. In the case of EMG data, we aim to perform amplitude and frequency analysis, co-contraction analysis, and the use of signal processing techniques for noise removal and frequency-domain analyses. Techniques such as maximum voluntary contraction [5, 6], motor unit action spectral analysis and pattern recognition contribute to a comprehensive understanding of muscle activity during ladder work. The time required for posture estimation and analysis using the head-mounted 360-degree camera and stationary 360-degree camera will be com-

pared to check the difference in rapidity. The accuracy of ART scores assigned for each of these camera settings would be compared against each other based on automated computer vision-based ART calculation.

5 CONCLUSION

The integration of 360-degree cameras and Microsoft HoloLens-driven AR for posture estimation in ladder-related work can significantly enhance occupational safety and task performance. Literature points to clear demand for prevention and frequency reduction of WMSD cases using practices for workforce well-being and health [19]. Being able to analyze workplace ergonomics is key to predicting short- and long-term WMSD risk. Incorporating AR dimensions into the analysis could potentially offer ergonomists a clearer understanding of posture due to the real-time overlay, reducing reliance on visual imagination. However, there are certain limitations to this proposed study that need to be addressed. First, this study focuses on specifically A-frame ladder use, and this may not be scalable to all types of work. Second, the orientation of the AR headset is up to the ergonomist's discretion, so might not be consistent between tasks. Lastly, there could be occlusions in case of equipment of other personnel blocking the line of sight for the workers, and the posture estimation result will be considered as the solution. Despite these limitations, the integrated system would promote a safer working environment by providing workers with real-time feedback, offering visual cues for self-correction and adherence to safety guidelines during ladder-related tasks. The proposed study currently proposes preprocessing the 360-degree imagery before analysis. There has been research conducted to directly process 360-degree video and imagery for more efficiency [26]. The head-mounted deployment of 360-degree cameras enables comprehensive views of the work environment without the need to actively pan camera views, ensuring more accurate multi-person posture estimation, including all workers in the work area.

The proposed solution marks a pioneering step towards advancing AR-enhanced occupational safety, paving the way for a seamless technology integration into workplace practices to ensure the well-being and efficiency of workers in elevated environments.

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